

JATIN KUMAR

Aspiring AI Engineer & Data Scientist

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Summary

CS graduate (Bachelors, VIT Vellore, India) beginning the M.Sc. Computer Science at Leibniz Universität Hannover in October 2026, resident in Hannover from 2nd October. Hands-on Python experience across machine learning, data cleaning, and pipeline development. Published first author at ICCS'25 (Springer, Singapore), applying statistical methods.

Work Experience

ML Research Intern

Sep 2024 – Apr 2025

IIT-BHU, Varanasi, IN

- Designed and ran controlled experiments across 3 structured molecular datasets, writing Python scripts to preprocess, clean, and validate raw data for accuracy before training.
- Applied statistical/time-series techniques (Mahalanobis distance, covariance modeling, Elastic Weight Consolidation) to reduce forgetting, achieving 90.89%–92.62% accuracy and 0.0–0.27 forgetting, outperforming 10+ SOTA methods.
- Documented methodology for peer review; published as first author at ICCS'25 (Springer LNCS) — [Paper](#)

Education

M.Sc in Computer Science

Oct 2026 – Present

Leibniz Universität Hannover, Hannover, Germany

B.Tech. in Computer Science and Engineering

Sep 2022 – Jul 2026

Vellore Institute of Technology (VIT), Vellore, India

Grade: 1.8

Relevant Courses: Machine Learning, Computer Vision, Artificial Intelligence, Statistics, Algorithms

Projects

1. CredCheck - Credit Risk Prediction

Python, scikit-learn, XGBoost, pandas, Streamlit

- Cleaned the German Credit dataset (1,000 applicants, 522 left after dropping incomplete rows), label-encoded 8 features and handled class imbalance with balanced class weights.
- Tuned 4 tree-based models with 5-fold grid search. XGBoost reached 69.5% test accuracy, up from 58.1% for a decision tree, and runs in a Streamlit app for live predictions. Code at: [CredCheck](#)

2. CommentLens - YouTube Viewer Sentiment Analyzer

Python, LightGBM, Hugging Face, MLflow, DVC, Flask, Docker, GCP Cloud Run, Chrome Extension

- Built a Chrome extension that labels a video's comments positive, neutral or negative and charts sentiment over time. It runs on a Flask API on Cloud Run that scales to zero, and deploys automatically from GitHub Actions without stored keys.
- Ran 7 tracked experiments in MLflow and DVC. Tuning LightGBM with 100 Optuna trials lifted accuracy from 65.1% to 78.7% and negative-class recall from 0.01 to 0.61. Switched production to a transformer that classifies 500 comments in about 16s on CPU. Code at: [CommentLens](#)

3. Aarogyam – AI-based Health and Disease Analyzer

Python, PyTorch, Gemini API, Streamlit

- Built a data pipeline ingesting real-time numerical sensor streams (heart rate, SpO2) from an IoT device, writing scripts to parse, filter, and validate incoming telemetry for downstream analysis.
- Integrated an LLM (Gemini API) via prompt engineering for symptom-based diagnostic reasoning over structured sensor data and unstructured text/image inputs, achieving under 2s inference latency. Delivered via a documented Streamlit UI. Live at: [Aarogyam](#)

Technical Skills

Languages: Python, C++, SQL

Machine Learning & Data: Scikit-learn, XGBoost, PyTorch, Pandas, NumPy, NLP, Feature Engineering, Data Cleaning, Model Evaluation (ROC-AUC, Precision/Recall)

AI/LLM: Hugging Face Transformers, Generative AI, Prompt Engineering, LangChain, RAG, Vector Databases (ChromaDB)

Software Engineering & Platforms: FastAPI, Git/GitHub, Jupyter, Streamlit, MySQL, AWS, Vercel

Languages

English: C1 (Advanced) **German:** A2 (Elementary)

Achievements

- Published Researcher, ICCS'25 (Springer LNCS), Singapore – conference paper on continual learning
- Volunteered as a fund-raiser for Muskurahat Foundation (A child-welfare organization)
- Top 10 Finalist, Yuvamanthan Hackathon 2024, organized by IIT Madras